

VIDYAVARDHAKA COLLEGE OF ENGINEERING

Autonomous Institute, Affiliated to Visvesvaraya Technological University, Belagavi

Gokulam, 3rd Stage, Mysuru 570 002

Sixth Semester B. E. Examinations

NUMERICAL METHODS AND ITS APPLICATIONS

Duration: 3 hours

Max. Marks: 100

INSTRUCTION TO STUDENTS

1) Answer one full question from each module.

Q. No.	Module-I	Marks	BL	CO												
1. (a)	Explain Inherent, Round-off and Truncation errors with an example. Also find the Absolute, Relative and percentage error in x_A when $x_T = \frac{1}{3}$ and $x_A = 0.333$.	10	2	1												
1. (b)	Extract a quadratic factor using Bairstow method from the equation $x^4 + 4x^3 - 7x^2 - 22x + 24 = 0$ with initial conditions $p_0 = 0.5$ and $q_0 = 0.5$.	10	3	2												
Module-II																
2.(a)	Obtain the solution of the system of equations $3x_1 + 3x_2 + 4x_3 = 5$; $2x_1 + x_2 + x_3 = 7$; $x_1 + 3x_2 + 5x_3 = 6$ using matrix partition method	10	4	2												
2.(b)	Solve the following system of equations by Cholesky method: $x_1 + 2x_2 + 3x_3 = 5$; $2x_1 + 8x_2 + 22x_3 = 6$; $3x_1 + 22x_2 + 82x_3 = -10$.	10	3	3												
Module-III																
3.(a)	Fit a first order derivative of the polynomial which takes the values <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>$f(x)$</td><td>0</td><td>10</td><td>38</td><td>96</td><td>196</td></tr></table>	x	1	2	3	4	5	$f(x)$	0	10	38	96	196	8	3	2
x	1	2	3	4	5											
$f(x)$	0	10	38	96	196											
3.(b)	Find the approximate value of the integral $I = \int_0^1 \frac{dx}{1+x^2}$ using Romberg's method.	12	4	3												
OR																
4.(a)	Find $\frac{dy}{dx}$ at $x = 1$ from the following table. <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>$f(x)$</td><td>1</td><td>3</td><td>7</td><td>13</td></tr></table>	x	0	1	2	3	$f(x)$	1	3	7	13	8	3	2		
x	0	1	2	3												
$f(x)$	1	3	7	13												
4.(b)	Evaluate the integral $I = \int_{y=1}^2 \int_{x=1}^2 \frac{dxdy}{x^2+y^2}$ using Simpson's rule with $h = k = 0.25$.	12	4	3												
Module-IV																
5.(a)	Using the Householder's transformation reduce the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ into a tridiagonal matrix.	8	3	2												

MODEL QUESTION PAPER : 2024-25

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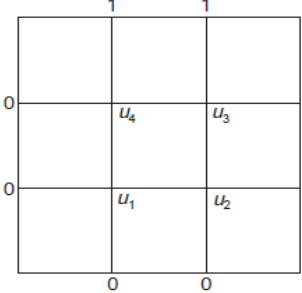
5.(b)	Using the Jacobi method, find all the eigenvalues of the matrix $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 2 \\ -1 & 2 & 1 \end{bmatrix}$. Iterate till the off-diagonal elements in magnitude are less than 0.0005.	12	3	2
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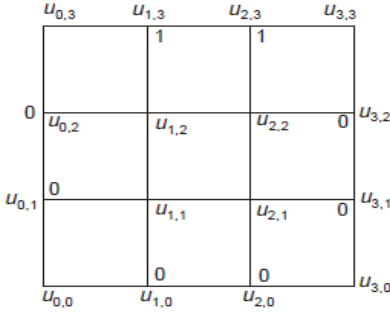
OR

6.(a)	Find all the eigenvalues of the symmetric matrix $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 1 \end{bmatrix}$ using Givens method.	8	3	2
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6.(b)	Find all the eigenvalues of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 1 & 3 & 2 \end{bmatrix}$ using Rutishauser method. Iterate till the elements of the lower triangular part are less than 0.05 in magnitude.	12	3	2
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Module- V

7.(a)	Solve the equation $u_{xx} + u_{yy} = 0$ in the domain of the figure given below by Successive Over Relaxation method. 	10	3	2
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7.(b)	Solve the equation $u_{xx} + u_{yy} = 0$ in the domain of the figure given below by Alternating Direction Implicit method. 	10	3	2
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